

OLUFEMI OSALONI

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SUMMARY

Results-driven Data Scientist with 9+ years of experience delivering enterprise-scale AI and machine learning solutions that generate measurable business value. Achieved \$7M+ in cost savings through predictive maintenance, inventory optimization, and intelligent automation projects across industrial operations. Currently extending this expertise into Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems to enhance analytics automation, customer intelligence, and marketing decision support. Proven success integrating explainable ML frameworks and scalable AI systems to drive data-informed business strategies.

CORE COMPETENCIES

- **Machine Learning & Generative AI:** Predictive Modeling, Generative AI (RAG, LangChain), Large Language Models (LLMs: OpenAI), Vector Databases (FAISS, ChromaDB), Time Series Forecasting, Deep Learning(LSTM, Autoencoders), Natural Language Processing (NLP), Computer Vision (CNNs, OpenCV), Clustering
- **MLOps & Cloud:** AWS (SageMaker, Glue, Lambda, S3), Azure (Azure Data Factory, AzureML), GCP (Familiar), SQL Server, Redshift, Hive, Neo4j, MLflow, DataRobot, CI/CD Pipelines (GitHub, Bitbucket, CircleCI), Docker, Git
- **Programming Languages:** Python, SQL, R, C/C++, PySpark, SAS, MATLAB
- **ML Frameworks:** Scikit-learn, Pandas, NumPy, SciPy, Statsmodels, XGBoost, TensorFlow, PyTorch, Keras, Pydantic-AI
- **NLP:** NLTK, spaCy, Sentence-BERT, HuggingFace Transformers
- **Data Visualization:** Power BI, Tableau, Plotly-Dash, Streamlit
- **Statistical & Experimental Methods:** Hypothesis Testing, Queuing Theory, A/B Testing, Experimental Design, Statistical Modeling
- **Data Engineering & ETL Pipelines:** Databricks, Alteryx, SQL Server, Redshift, Hive, SAP ERP
- **Modeling, Analytics & Methodologies:** Feature Engineering, Digital Twins, Weibull Analysis, FMEA, Survival Modeling, Anomaly Detection, Exploratory Data Analysis (EDA), Geospatial Analytics, Simulation Modeling, Process Optimization, Model Lifecycle Management (Tuning, Automated Retraining, Monitoring), Agile Development
- **Applied Use Cases:** Customer Lifetime Value (CLV), Churn Prediction, Customer Segmentation, Propensity Modeling, Recommendation Systems, Pricing Optimization

EDUCATION

- M.S in Analytics – Villanova University School of Business | May 2018 | GPA: 4.0/4.0
- Ph.D. in Electrical Engineering (ABD) – Tennessee Technological University | Dec 2005
- M.S in Electrical Engineering – Tennessee Technological University | Aug 2003
- B.S in Electrical and Electronic Engineering – Obafemi Awolowo University | Dec 1998

WORK EXPERIENCE

Senior Data Scientist – LyondellBasell Industries

Nov 2021 – Present / Houston, TX

- Analyzed equipment performance data and identified key failure patterns through statistical modeling and anomaly detection using Python, enabling predictive maintenance strategy that reduced downtime by 25% and delivered \$2M in cost savings through data-driven operational insights.
- Led cross-functional initiatives and collaborated with internal business units and external stakeholders to translate advanced ML solutions into measurable business impact, mirroring a consulting engagement model.
- Built automated analytics pipeline for product-shipping optimization, leveraging advanced ML techniques to reduce manual analysis time by 90% and improve decision-making speed for operations team.
- Mentored junior data scientists in model development, MLOps, and technical documentation best practices.
- Engineered an enterprise-scale global energy-management analytics platform on Databricks, integrating multi-source streaming data from industrial historians (OSI PI, Aspen IP21) and delivering interactive Power BI dashboards for real-time monitoring and operational optimization across global operations.

Data Scientist – TechnipFMC

Sep 2016 – Nov 2021 / Houston, TX

- Developed PowerBI dashboard for equipment reliability, integrating MTBF tracking, Weibull analysis, and FMEA to reduce unplanned downtime by 25% and save \$500K annually.
- Delivered \$5M+ cost savings by developing an end-to-end machine learning recommendation system leveraging Sentence-BERT embeddings to match and suggest similar part descriptions based on textual similarity(excess inventory).
- Developed regression and forecasting models to support production order lead-time prediction and pricing strategy optimization which reduced time-to-market by 30% and increase market share of subsea hydraulic chokes.
- Led design and implementation of AWS cloud infrastructure (SageMaker, S3, Lambda, Glue), driving a 50 % increase in data-processing speed and enterprise-wide scalability across three departments.

Electronics and Power Systems Engineer – FMC Technologies

Sep 2012 – Sep 2016 / Houston, TX

- Led innovation initiatives in subsea power systems, identifying emerging industry trends and disruptive technologies
- Implemented conceptual engineering designs for Subsea Power System Networks using advanced simulation tools and MATLAB/Simulink

Senior Product Development Engineer – Eaton Corporation

Aug 2006 – Sep 2012 | Milwaukee, WI

- Designed and implemented commercial and utility-scale Grid-Tie Solar Inverters up to 500kW capacity
- Executed comprehensive qualification testing according to UL1741 and IEEE1547 industry standards

TRAININGS

- Master RAG: Ultimate Retrieval-Augmented Generation
- Career Essentials in Generative AI – Microsoft/LinkedIn (ongoing)
- The Complete Agentic AI Engineering Course – design, orchestrate, and deploy robust AI agents and agentic workflows (ongoing)
- Advanced LangChain, Vector Databases, and Prompt Engineering Certification (ongoing)

AI/ML PROJECTS

- Designed and implemented a Retrieval-Augmented Generation (RAG) prototype leveraging LangChain, LangSmith, LangGraph, vector databases (FAISS/Chroma), and LLMs, with a Streamlit-based user interface, to automate document search and answer generation, improving retrieval speed and accuracy for internal knowledge workflows.
- Developed a mock healthcare analytics pipeline integrating EMR-like structured ERP data to simulate patient flow and forecast resource utilization, leveraging time-series forecasting and Tableau dashboards for clinical decision support.
- Developed and backtested algorithmic trading strategies using time series analysis and forecasting modeling on stock price data; leveraged Python (Pandas, NumPy, Statsmodels) to generate quantitative signals from historical patterns.